

Final Project: Drug Interaction Informant

IAT 360

Group 12

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Repository (Datasets in Repository):

https://github.com/eyoung-15/IAT_360_Final_Project

Chosen Problem and its Significance:

The purpose of our project is to provide a chatbot for users to ask questions regarding whether or not they can safely take two different medications together. Our model responds to the user's query by answering whether the medication combo is safe, along with any potential medication interactions. This chatbot will help users start meaningful conversations with their medical provider. Users will learn about medications they are taking and medications they are interested in starting. We also hope that this chatbot sparks general curiosity for users about what medications they are putting in their bodies.

Model Used:

We used the LLAMA 2 model for our project. LLAMA 2 is a large language model. LLAMA 2 is developed by Meta and is optimized for use cases involving dialogue. It is an auto-regressive language model and uses a transformer architecture. The model takes text input and generates text only. LLAMA 2 is trained on 2 trillion tokens of publicly available online data and is intended for use in English. This model was evaluated on several standard academic benchmarks:

- Commonsense Reasoning: 63.9
- World Knowledge: 48.9
- Reading Comprehension: 61.3
- Math: 14.6
- MMLU (Massive Multitask Language Understanding): 45.3
- BBH (BIG-Bench Hard): 32.6
- AGI Evaluation: 29.3
- TruthfulQA (truthfulness): 33.29
- Toxigen (distinguishing between toxic statements): 21.25

Because this model deals with sensitive medical information and provides responses that can lead to potentially fatal outcomes, we have decided not to deploy this model and therefore have not requested a licence to share and deploy this model.

Users/Use Cases:

The users of our model will be anyone interested in starting medications, those who currently are taking medications, and individuals who have general curiosity and interests in pharmaceuticals. We hope that rather than a replacement for a licensed medical practitioner, this chatbot may spark a meaningful medical conversation with their practitioner.

Users would typically use this model on their home computer or personal mobile phone. A user may use this chatbot by asking about new medications they are curious about starting to see if they may interact with any currently prescribed medications. Another user may ask about an over the counter drug they are unsure if they can take with their current medications. Another user may want to double check their current prescribed medications to understand any potential interactions better.

The data required for this model includes a user query with two medications for the chatbot to compare.

The Datasets

The model is trained on a dataset from Kaggle, called “Drug-Drug Interactions”. The dataset contains three columns. Column 1 is the first generic drug name, column 2 is the second generic drug name, Column 3 is the drug interaction description. The data set contains more than 1600 unique drugs. Unfortunately, the dataset only contains the generic drug names and not the brand names.

To solve the problem of brand name detection, we found another Kaggle dataset called “Drugs, Side Effects and Medical Condition”. This dataset contains seventeen columns, but we only utilized the drug_name and generic_name column. We used the two columns in the dataset to create a dictionary that would match any brand names mentioned in a user’s query to their actual generic name. That way the chatbot recognizes drugs by their brand name, when previously the chatbot would have only been able to recognize drugs by their generic name. There are 2912 drugs in this dataset.

Both datasets show bias towards drugs and medications in the USA, as the first dataset is derived from Stanford University Medicine and the second dataset contains information from drugs.com. Because we are not medical professionals, we did not change or alter any of the data in these datasets. What is missing from this data is information from other regions.

Guardrails

Because this model contains medical information, we added a few guardrails to ensure maximum safety.

We added a guardrail that provides a disclaimer at the end of each response, which states:

“This chatbot should not be used in replacement of a licensed medical practitioner, as it may contain inaccuracies. Please always consult your medical practitioner before starting and/or stopping any medications.”

We also added a guardrail that asks users to include two drug names in their query if no medications were detected in their previous query. We implemented this by telling the chatbot to return this message if the returned similarity score is less than 0.9. The guardrail message states:

“Please re-enter two drug names so that I can check whether or not they can be safely taken together. If you had entered two drug names, I could not find information about this drug combination.”

Making the Model Application Ready

We did not make any changes to the model, but embedded our datasets to add to its knowledge base. To train the model on the drug interaction dataset we only embedded the rows of two drug names, and did not embed the "Interaction Description" column. This is because we only want to search for drug names in user queries and match them to any drugs in our interaction database. However, we still used the "Interaction Description" column as a part of the context.

Receiving Data From the User

Users will input a query about whether or not they can take two medications together. For example, "Can I take Prozac and Lexapro together?" The data we are collecting from users are the queries they ask the chatbot. Each time the user enters a query, the query is cleaned so only drug names are extracted from the query. That way the model only cares about the relevant key words. The model has a function that extracts drug names from the user query and then adds them to a list. If a brand name is detected, the function converts it to its generic name and then adds it to the list. This list is used for the model to match the drugs to the database.

If this model were to be deployed for usage, user information would be stored temporarily in memory (the computer's RAM) for the session. Information would be cleared from memory upon the session ending. Because this model deals with private medical information, it would not retain users' queries after the session ends. This process would occur for each user in all circumstances in order to protect sensitive medical information.

Challenges, Ethical Issues, Biases

Since our model provides medical information, we encountered ethical issues regarding responses leading to potentially fatal outcomes. Users should be made aware that they cannot rely solely on information provided by the chatbot. Therefore, we addressed this ethical issue by adding guardrails to our model. These guardrails are meant to keep users informed on the model's purpose. The guardrails ensure users know that the model has potential to be incorrect, which is why they should always contact a doctor before stopping or starting medications. It tells users that the responses from the chatbot are never to be used as a replacement for a licensed medical practitioner. As well, if users input something not within the dataset, the model asks users to re-input their query and that their original input may not be within the database. Our model is limited to being a supplementary companion.

A user issue we noticed early on was that the model would only be able to retrieve common names of drugs, rather than their brand names. To overcome this, we implemented another dataset mapping drug names to their brand names. This made our model more usable, as users can inquire about both common and brand names of their medications.

Another bias issue with our model is that our datasets are both American. This means that our data is biased to include American drugs and medications, as well as American brand names of drugs and medications. Therefore, a case our model would fail in is providing drug interaction

information in other regions. A way to address this issue if the model were to be expanded is to include datasets for other regions so that the model is not limited to use in the USA.

References

Drug-Drug Interactions Dataset:

<https://www.kaggle.com/datasets/mghobashy/drug-drug-interactions>

Drugs, Side Effects and Medical Condition Dataset:

https://www.kaggle.com/datasets/jithinanievarghese/drugs-side-effects-and-medical-condition?select=drugs_side_effects_drugs_com.csv

LLAMA 2 Model through Hugging Face

<https://huggingface.co/meta-llama/Llama-2-7b-chat-hf>